

Assignment 7: Memory Management

Q1. Implement the three classic strategies for placing a process into memory holes. Given a list of available memory blocks (holes) and a list of process requirements, write a program to simulate:

1. **First Fit:** Allocate the process to the first hole that is big enough.
2. **Best Fit:** Allocate to the smallest hole that is big enough (to minimize wasted space).
3. **Worst Fit:** Allocate to the largest hole available (to leave large remaining chunks).

Input:

- Block sizes: [100K, 500K, 200K, 300K, 600K]
- Process sizes: [212K, 417K, 112K, 426K]

Output: A mapping of which process is in which block and the total Fragmentation (leftover space).

Q2. Simulate the MMU (Memory Management Unit) hardware logic. Write a program that mimics a **Two-Level Paging System**.

- **Virtual Address:** 32-bit.
 - Outer Page Table: 10 bits.
 - Inner Page Table: 10 bits.
 - Offset: 12 bits.
- 1. Create a "Dummy" Page directory (outer page table) and inner Page Tables.
- 2. Given a hexadecimal Virtual Address (e.g., 0x12345678), your program must extract the Directory Index, Table Index, and Offset.
- 3. "Look up" the physical frame (just return a simulated value) and calculate the final Physical Address.